

Tensegrity Rover Simulation Report

1. Overview of the Codebase

The codebase is a browser-based 3D physics simulation of a six-strut, 24-cable tensegrity rover. The rover is designed to navigate through a simulated Martian terrain, complete with rough sand, stones, obstacles, and changing gravity constraints. The simulation employs a custom physics engine running at 500 Hz to handle rigid-body mechanics, spring/damper cables, constraints, and collisions.

The code features an **Additive Drive Controller Suite**, imported from MATLAB models, exposing various control strategies for rolling the tensegrity robot dynamically. The suite includes implementations of Linear Quadratic Regulators (LQR), Iterative LQR, Model Predictive Control (MPC) using Quadratic Programming (QP), and Neural Network-based geometric policies.

2. Controller Diff: Which is Better and Why?

We analyzed and ran tests across the 8 implemented controllers. Below is a breakdown of their strengths and why certain ones perform better under specific conditions.

- **LQR & Central Payload LQR (Fast & Reactive):**
 - **Pros:** Extremely fast to compute, very stable near equilibrium points.
 - **Cons:** Does not account for physical limits (e.g., maximum cable tension).
 - **Verdict:** Best for flat terrains where the vehicle is close to its nominal state.
- **iLQR & Variants (Minimax & Input Penalty):**
 - **Pros:** Handles the non-linear dynamics of the tensegrity structure much better by linearizing around a trajectory. The Minimax variant adds robustness against worst-case disturbances.
 - **Cons:** Can get stuck in local minima during its line-search phase; higher computational cost.
 - **Verdict:** Best for unpredictable terrains where disturbances (rocks, uneven surfaces) require robust non-linear trajectory adjustments.
- **QP-MPC & Central Payload QP-MPC (Constrained & Optimal):**
 - **Pros:** Explicitly models hardware constraints (max tension, actuator limits). Re-plans continuously to avoid constraint violations.
 - **Cons:** High computational latency (requires running a convex optimization step at every control tick).
 - **Verdict:** **Overall Best Controller** for ensuring hardware safety (e.g., not snapping cables or crushing the payload) while maintaining a strict path. The Central Payload version optimally dampens core shocks while navigating.
- **Neural Network Controller (Fast Heuristics):**
 - **Pros:** Runs inference in $O(1)$ time without solving Riccati equations or QPs.
 - **Cons:** Performance heavily depends on training data; deterministic fallback used here works fine but isn't globally optimal.

3. Combining Controllers into Groups

The controllers can be combined into logical groups based on their mathematical paradigms:

- **Group 1: Linear Feedback Control**
 - `lqr` : Standard LQR
 - `lqr_payload` : LQR tuned for central payload stability
- **Group 2: Iterative Non-Linear Control**
 - `ilqr` : Standard iLQR
 - `ilqr_minimax` : Robust Minimax iLQR (saddle-point)
 - `ilqr_minimax_penalty` : Minimax iLQR with explicit input change penalties
- **Group 3: Receding Horizon Constrained Control (MPC)**
 - `qp_mpc` : Box-constrained Receding-Horizon QP-MPC
 - `qp_mpc_payload` : QP-MPC optimized for payload state weighting
- **Group 4: Data-Driven Control**
 - `neural` : Neural Network Rolling Direction
- **Group 5: Natural Support**
 - `natural_support_face` : Default fall-back unactuated geometry policy
- **Group 6: All Controllers (Ensemble Mode)**
 - This mode would evaluate the cost functions of all active controllers in parallel and pick the action from the one with the lowest predicted cost-to-go, ensuring safety and performance across all scenarios.

4. Testing in Terrain 10

Level 10 is the **Central Payload Core G-Force Isolation** test environment.

- **Objective:** Navigate a 1.6 kg central payload over 10 staggered spherical hazards on rough Martian terrain.
- **Condition:** Must maintain a nominal speed of 0.20 m/s without the internal payload experiencing more than 1.5 Gs of proper acceleration or travelling more than 75 mm radially.
- **Result:** The simulation rigorously tests the rover's suspension cables. Shocks generated by impacts with the 10 obstacles are absorbed by the tension-only spring/damper suspension, allowing the outer cage to bounce heavily while the inner core remains isolated and below the 1.5 G threshold.